4.07 Hyperbolic Functions				
4.07a	Definitions		a) Understand and be able to use the definitions of the hyperbolic functions $\sinh x$, $\cosh x$ and $\tanh x$, in terms of exponentials. *Including the domain and range of each function.*	H1
4.07b			b) Know and be able to sketch the graphs of the hyperbolic functions.	
4.07c			c) Know and be able to use the identity $\cosh^2 x - \sinh^2 x \equiv 1$. *Learners may be asked to derive or use other identities, but no prior knowledge of them is assumed.* [*Prior knowledge of other identities is excluded.*]	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.07d	Differentiation and integration		d) Be able to differentiate and integrate hyperbolic functions.	H2
4.07e 4.07f	Inverse hyperbolic functions		e) Understand and be able to use the definitions of the inverse hyperbolic functions and their domains and ranges. f) Be able to derive and use expressions in terms of logarithms for the inverse hyperbolic functions. *Includes the notation:* $\sinh^{-1} x$, $\cosh^{-1} x$, $\tanh^{-1} x$ *and* $\text{arsinh}\, x$, $\text{arcosh}\, x$, $\text{artanh}\, x$.	H3 H4